ISLAMIC UNIVERSITY OF TECHNOLOGY

Computer Science and Engineering (CSE)

Chem 4241: Chemistry

Periodic Table

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Pauli exclusion principle

The **Pauli exclusion principle** states that *no two electrons in an atom can have the same four quantum numbers.*

Consider an atom has two electron. If one electron has the quantum numbers of n = 1, l = 0, m = 0, and s = +1/2, then, other electron must has n = 1, l = 0, m = 0, and s = -1/2.

Each subshell holds a maximum of twice as many electrons as the number of orbitals in the subshell. Thus, a 2*p* subshell, which has three orbitals (with l = -1, 0, and 1), can hold a maximum of six electrons. The maximum number of electrons in various subshells is given in the following table.

Subshell	Number of orbitals	Maximum number of electron
s (l = 0)	1	2
p (l = 1)	3	6
d (l = 2)	5	10
f (l = 3)	7	14

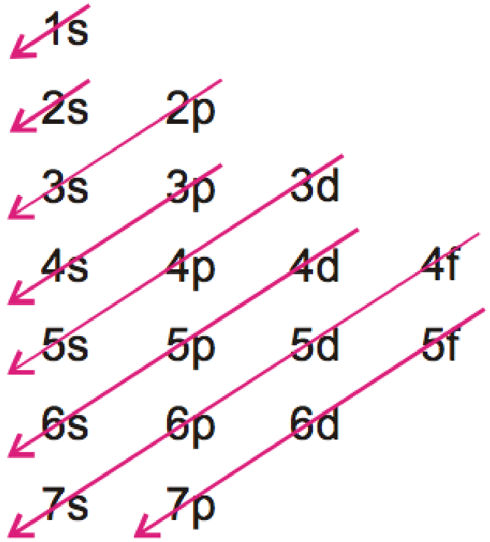
Building Up Principle or Aufbau Principle

In the ground state of an atom, the electrons tend to occupy the available orbitals in the increasing order of energies, the orbitals of lower energy being filled first.

The energy of an orbital is determined by the sum of principal quantum number (n) and the azimuthal quantum number (l). This rule is called **($n + l$)** rule**.** There are two parts of this rule :

(*a*) The orbitals with the lower value of ($n + l$) has lower energy than the orbitals of higher ($n + l$) value. For example, let us compare the ($n + l$) value for 3*p* and 3*d* orbitals. For 3*p* orbital $n = 3$, $l = 1$ and $n + l = 4$ and for 3d orbital $n = 3$, $l = 2$ and $n + l = 5$. Therefore, 3p orbital is filled before 3*d* orbital.

(*b*) When two orbitals have same ($n + l$) value, the orbital with lower value of n has lower energy. Similarly, for 4*p* and 5*s* orbitals, the ($n + l$) values are (4 + 1) and (5 + 0) respectively. In this case 4*p* orbital has lesser value of n and hence it has lower energy than 5*d* orbital and is filled first.

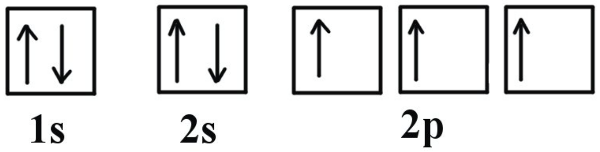


Hund's Rule (Maximum no of electrons remains in unpaired state)

1. Every orbital in a sublevel is singly occupied before any orbital is doubly occupied.
2. All of the electrons in singly occupied orbitals have the same spin (to maximize total spin).

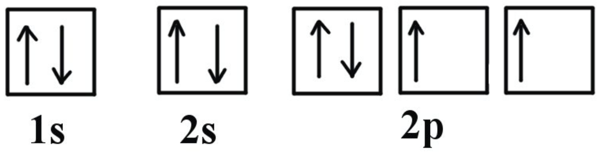
Consider the correct electron configuration of the nitrogen (Z = 7) atom:

$$1s^2\ 2s^2\ 2p^3$$



The p orbitals are half-filled; there are three electrons and three p orbitals. This is because the three electrons in the 2p subshell will fill all the empty orbitals first before pairing with electrons in them.

Consider oxygen (Z = 8) atom, the element after nitrogen in the same period; its electron configuration is:



Oxygen has one more electron than nitrogen; as the orbitals are all half-filled, the new electron must pair up.

Dobereiner's Triads

A tabular arrangement of elements in rows and columns, highlighting the regular repetition of properties of the elements, is called a periodic table.

Johann Wolfgang Dobereiner

Dobereiner's Triads

- **Triads** Groups of three elements which showed similar properties
- Atomic mass of the middle element is approximately the mean of the atomic masses of other two elements

In the triad of Li, Na and K the atomic mass of Na (23) is the mean of the atomic masses of Li and K

6.9 + 39 = 45.9 ÷ 2 = 22.95

Element	**Atomic mass**
Li	6.9
Na	**23**
K	39

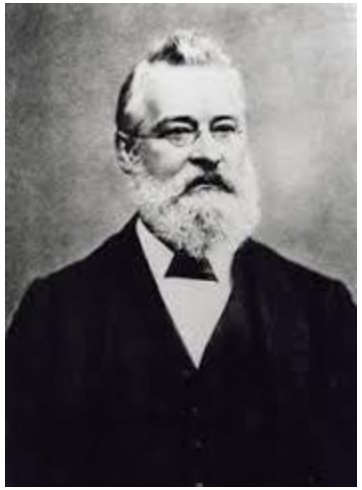
Dobereiner's Triads

Element	Atomic mass
Li	6.9
Na	**23**
K	39
Ca	40.1
Sr	**87.6**
Ba	137.3
Cl	35.5
Br	**79.9**
I	126.9
S	32
Se	**79**
Te	128

Features

- Only a few triads could be identified
- System of triads could not continue

Newland's Octaves



John Alexander Reina Newlands

- Fifty-six elements were discovered.
- Newlands arranged them in the increasing order of their atomic masses.
- Every eighth element had properties similar to the first.

Newlands' Octaves

H	Li	Be	B	C	N	O
F	Na	Mg	Al	Si	P	S
Cl	K	Ca	Cr	Ti	Mn	Fe
Co, Ni	Cu	Zn	Y	In	As	Se
Br	Rb	Sr	Ce, La	Zr	Di, Mo	Ro, Ru
Pd	Ag	Cd	U	Sn	Sb	I
Te	Cs	Ba, V	Ta	W	Nb	Au
Pt, Ir	Os	Hg	Tl	Pb	Bi	Th

Newland’s Octaves

Features

- Out of the 56 elements, elements up to Ca could be arranged.
- After Ca every eighth element did not possess properties similar to the first.
- To fit the existing elements two elements were placed in the same position which differed in their properties.
- Inert (noble) gases were not included because they were not discovered.

Mendeleev’s Periodic Table

Dmitri Ivanovich Mendeleev

- Examined the relationship between the atomic masses of elements and their physical and chemical properties.
- Believed that atomic mass was the most fundamental property in classifying the elements

Mendeleev’s Periodic Law

The physical and chemical properties of elements are a periodic function of their atomic masses

Features

Periods

- Horizontal rows, numbered **1 to 7**
- Properties of elements in a period show regular gradation from left to right

Groups

- Vertical columns, numbered **I to VIII**.

I	II	III	IV	V	VI	VII	VIII		
H 1.01									
Li 6.94	Be 9.01	B 10.8	C 12.0	N 14.0	O 16.0	F 19.0			
Na 23.0	Mg 24.3	Al 27.0	Si 28.1	P 31.0	S 32.1	Cl 35.5			
K 39.1	Ca 40.1		Ti 47.9	V 50.9	Cr 52.0	Mn 54.9	Fe 55.9	Co 58.9	Ni 58.7
Cu 63.5	Zn 65.4			As 74.9	Se 79.0	Br 79.9			
Rb 85.5	Sr 87.6	Y 88.9	Zr 91.2	Nb 92.9	Mo 95.9		Ru 101	Rh 103	Pd 106
Ag 108	Cd 112	In 115	Sn 119	Sb 122	Te 128	I 127			
Ce 133	Ba 137	La 139		Ta 181	W 184		Os 194	Ir 192	Pt 195
Au 197	Hg 201	Ti 204	Pb 207	Bi 209					
			Th 232		U 238				

Mendeleev’s Periodic Table

Demerits

- H resembles alkali metals and halogens. No fixed position could be given to H.
- Isotopes of same elements have different atomic masses. Each of them should be given a different position. As isotopes are chemically similar, they were given same position.
- Co with higher atomic mass (58.93) is placed before Ni (58.71).
- Mn is placed with halogens which totally differ in the properties.

Merits

- Successful classification of all known elements
- Some vacant/ blank spaces were left for elements yet to be discovered.
- Mendeleev predicted properties of these elements even before they were discovered. Later they were found to be correct.
- Noble gases were discovered later and placed in the table without disturbing the positions of other elements.

Modern Periodic Table

Henry Gwyn Jeffreys Moseley

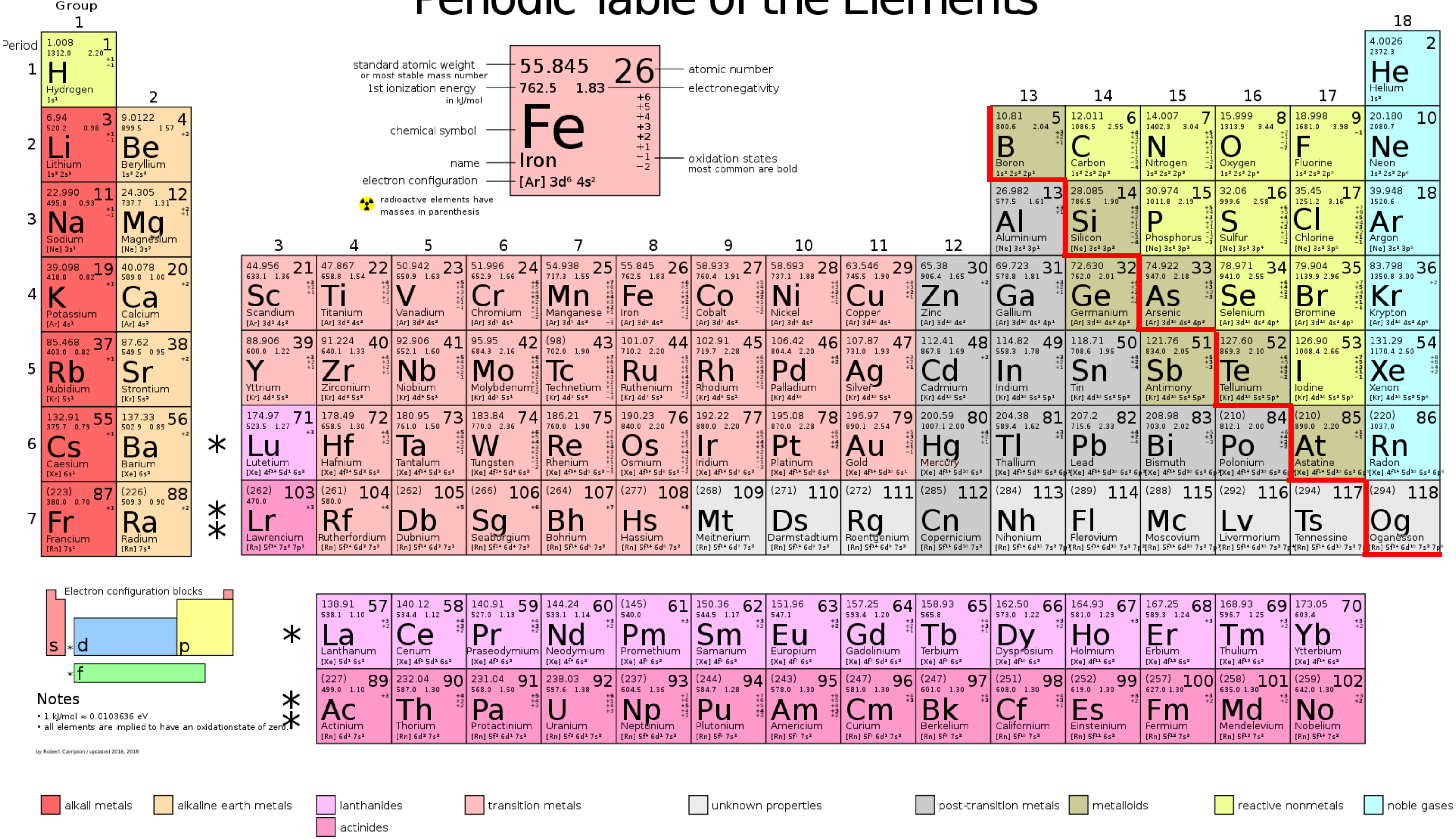
Atomic number is the most fundamental property of an element and not its atomic mass – **Henry Moseley**

Atomic number (Z) = number of protons in the nucleus of the atom
Also represents number of electrons in the outer shell

Mendeleev's Periodic Law was modified into **Modern Periodic Law:**
The chemical and physical properties of elements are a periodic function of their atomic numbers

Not much different from Mendeleev's Periodic Table

Periodic Table of the Elements



Modern Periodic Table

Features

- **Periods** (number of shells)
- Horizontal rows, Seven, numbered 1-7
- Elements in the same period have same number of shells which is equal to the period number. e.g. Period 7 has 7 Shells
- In each period a new shell starts filling up

- **Groups** (number of valence electrons)
- Vertical columns, Eighteen, numbered 1-18
- Elements in the same group have
  - Same number of valence electrons/ same outer electronic configuration,
  - Show same chemical properties
- Group 1: alkali metals
- Group 2: alkaline earth metals
- Group 17: halogens
- Group 18: inert/ noble gases

Modern Periodic Table

Metals – left hand side

Non-metals – right hand side

Main group elements – Groups 1, 2 and Groups 13-17.

Transition elements – Groups 3-12.

Inert gases – Group 18. Outermost shell contains 8 electrons

Inner transition – at the bottom, contain two series, viz. lanthanides, actinides

Lanthanides (Ce – Lu) – 14 elements, atomic numbers 58-71. Placed along with La (57), Group 3, Period 6. Close resemblance in properties to La

Actinides (Th – Lr) – 14 elements, atomic numbers 90-103. Placed along with Ac (89), Group 3, Period 7. Close resemblance in properties to Ac

Modern Periodic Table

Elements can be classified into four categories according to their electron configurations.

1.The noble gases. *These are elements in which the outermost s and p sublevels are filled.*

The noble gases belong to Group 0. The elements in this group are sometimes called the inert gases because they do not participate in many chemical reactions. The electron configurations for the first four noble-gas elements are listed below.

Helium	$1s^2$
Neon	$1s^2 2s^2 2p^6$
Argon	$1s^2 2s^2 2p^6 3s^2 3p^6$
Krypton	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

Notice that all of these elements have filled **outermost** *s* and *p* sublevels

Modern Periodic Table

2. The representative elements. *In these elements, the outermost s and p sublevel is only partially filled.*

The representative elements are usually called the Group A elements or main group elements.

For any representative element, the group number equals the number of electrons in the outermost energy level.

For example, the elements in Group 1A (lithium, sodium, etc.) have one electron in the outermost energy level.

Lithium	$1s^2 2s^1$
Sodium	$1s^2 2s^2 2p^6 3s^1$
Potassium	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

Carbon, silicon, and germanium, in Group 4A, have four electrons in the outermost energy level.

Carbon	$1s^2 2s^2 2p^2$
Silicon	$1s^2 2s^2 2p^6 3s^2 3p^2$
Germanium	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^2$

Modern Periodic Table

3. The transition metals. *These are metallic elements in which the outermost s sublevel and nearby d sublevel contains electrons.*

The transition elements, called Group B elements, are characterized by addition of electron to the *d* orbitals.

Scandium $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$

Yttrium $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^1$

4. The inner transition metals. *These are metallic elements in which the outermost s sublevel and nearby f sublevel generally contain electrons.*

The inner transition metals are characterized by the filling of *f* orbitals.

Cerium $[Xe]6s^2 5d^1 4f^1$

Thorium $[Rn]7s^2 6d^1 5f^1$

Modern Periodic Table

Merits

- Modern Periodic Table (atomic number) versus Mendeleev's Periodic Table (atomic mass)
- All isotopes of the same elements have different masses but same atomic number and occupy the same position
- Anomaly regarding Co (27) and Ni (28) disappears
- Classification of elements into blocks based on their electronic configuration

Finding Location of Elements in Periodic Table

a) Finding Period of Elements:

Period of the element is equal to highest energy level of electrons or principal quantum number.

For example:

$_{16}$**S:** $1s^2 2s^2 2p^6 \underline{\mathbf{3}}s^2 \underline{\mathbf{3}}p^4$ 3 is the highest energy level of electrons or principal quantum number. Thus period of S is 3.

$_{23}$**Cr:** $1s^2 2s^2 2p^6 3s^2 3p^6 \underline{\mathbf{4}}s^2 3d^4$ 4 is the highest energy level of electrons or principal quantum number. Thus period of Cr is 4.

b) Finding Group of Elements:

Group of element is equal to number of valence electrons of element or number of electrons in the highest energy level of elements. Another way of finding group of element is looking at sub shells. If last sub shell of electron configuration is "s" or "p", then group becomes A.

$_{19}$**K:** $1s^2 2s^2 2p^6 3s^2 3p^6 4\underline{\mathbf{s}}^1$ Since last sub shell is "s" group of K is A.

$_{35}$**Br:** $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4\underline{\mathbf{p}}^5$ Since last sub shell is "p" group of Br is A.

Elements in group B have electron configuration **ns** and **(n-1)d**, total number of electrons in these orbitals gives us group of element. Look at following examples.

$_{26}$Fe: $1s^2 2s^2 2p^6 3s^2 3p^6 \underline{\mathbf{4s^2 3d^6}}$ 6+2=8 B group

Finding Location of Elements in Periodic Table

Here are some clues for you to find group number of elements.

<u>Last Orbital-Group</u>

ns^1 1A

ns^2 2A

ns^2np^1 3A

ns^2np^2 4A

ns^2np^3 5A

ns^2np^4 6A

ns^2np^5 7A

ns^2np^6 8A

<u>Last Orbital-Group</u>

$ns^2(n\text{-}1)d^1$ 3B

$ns^2(n\text{-}1)d^2$ 4B

$ns^2(n\text{-}1)d^3$ 5B

$ns^2(n\text{-}1)d^4$ or $ns^1(n\text{-}1)d^5$ 6B

$ns^2(n\text{-}1)d^5$ 7B

<u>Last Orbital-Group</u>

$ns^2(n\text{-}1)d^6$ 8B

$ns^2(n\text{-}1)d^7$ 8B

$ns^2(n\text{-}1)d^8$ 8B

$ns^2(n\text{-}1)d^9$ or $ns^1(n\text{-}1)d^{10}$ 1B

$ns^2(n\text{-}1)d^{10}$ 2B

Example: Find period and group of $_{16}X$.

$_{16}X$: $1s^22s^22p^63s^23p^4$

Period: 3 and group: 2+4 = 6A

Example: Find period and group of $_{24}X$.

$_{24}X$:$1s^22s^22p^63s^23p^64s^23d^4$

Period: 4 and group: 4+2 = 6B

Practice Problems

Which of the following orbital diagrams or electron configurations are possible and which are impossible, according to the Pauli exclusion principle? Explain.

a. (↑↓) (↑↓) (↑)( )( )
1s 2s 2p

b. (↑↓) (↑↓↑) ( )( )( )
1s 2s 2p

c. (↑↓) (↑) (↑↑)( )( )
1s 2s 2p

d. $1s^3 2s^1$

e. $1s^2 2s^1 2p^7$

f. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$

Choose the electron configurations that are possible from among the following. Explain why the others are impossible.

a. $1s^2 2s^1 2p^6$
b. $1s^2 2s^2 2p^8$
c. $1s^2 2s^3 2p^6 3s^2 3p^6 3d^7$
d. $1s^2 2s^2 2p^6 3s^1 3d^9$

Group 3

Thallium has the ground-state configuration $[Xe]4f^{14}5d^{10}6s^2 6p^1$. Give the group and period for this element. Classify it as a main-group, a d-transition, or an f-transition element.

Write the orbital diagram for the ground state of cobalt. The electron configuration is $[Ar]3d^7 4s^2$.

52Te

The ground-state electron configuration of an atom is $1s^2 2s^2 2p^6 3s^2 3p^4$. What is the valence-shell configuration of the atom in the same group, but in Period 5?

Practice Problems

Use the building-up principle to obtain the configuration for the ground state of the gallium atom (Z = 31). Give the configuration in complete form. What is the valence-shell configuration?

Write an orbital diagram for the ground state of the iron atom.

What kind of subshell is being filled in Groups IA and IIA? in Groups IIIA to VIIIA? in the transition elements? in the lanthanides and actinides?

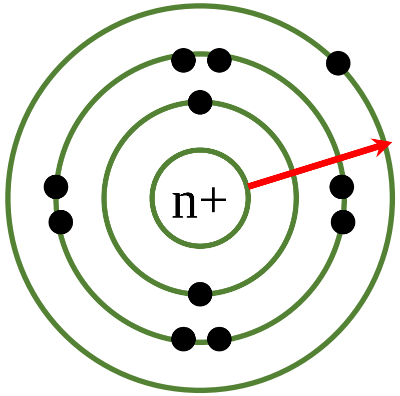
Bromine is a Group VIIA element in Period 4. Deduce the valence-shell configuration of bromine.

Write an orbital diagram for the ground state of the calcium atom. Is the atomic substance diamagnetic or paramagnetic?

Ans: Paramagnetic

Atomic Radius

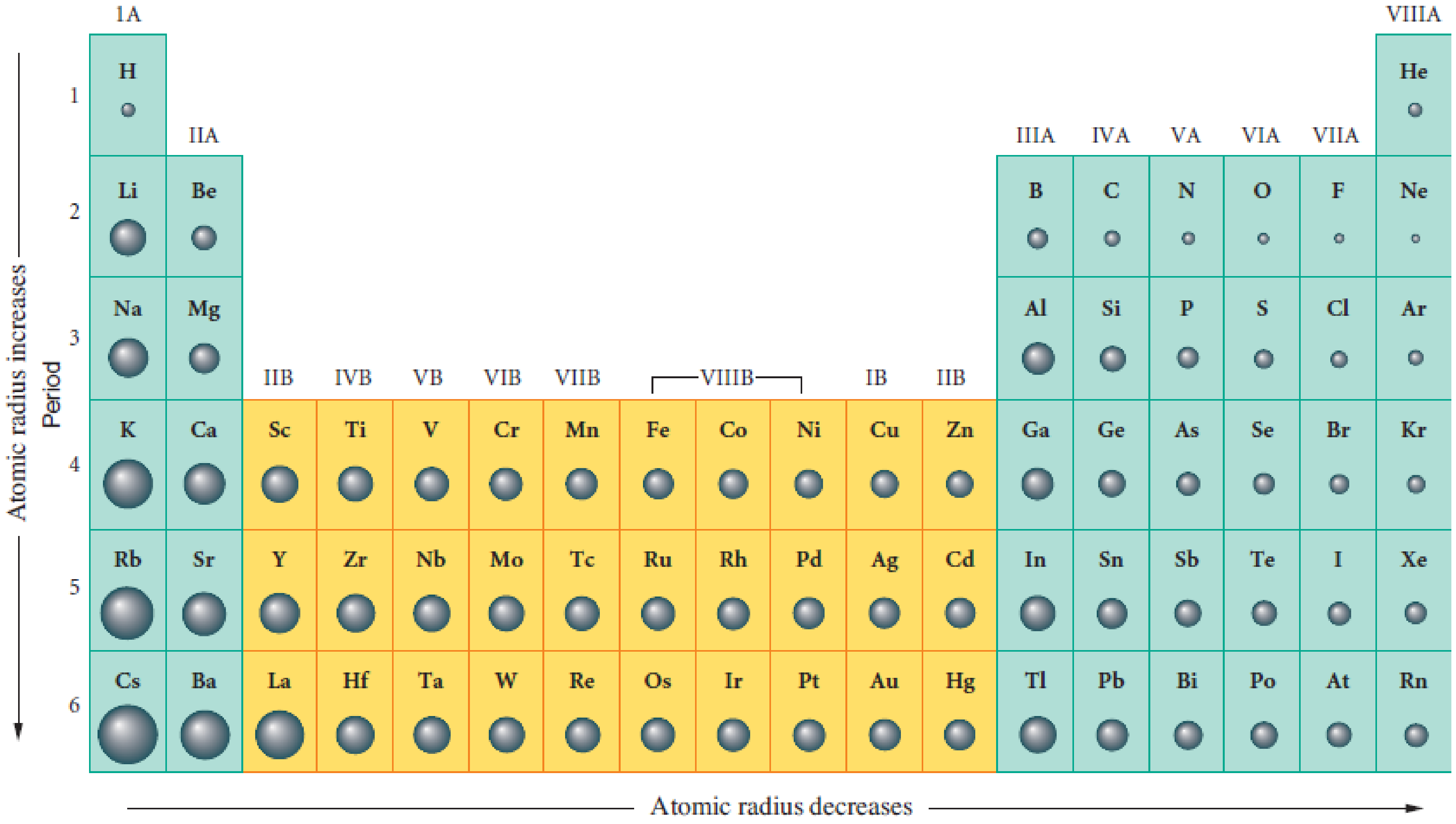
Atomic radius is the distance between the center of atom and the outermost shell.



General trends in size of atomic radii

- Within each period (horizontal row), the atomic radius tends to decrease with increasing atomic number (nuclear charge). The largest atom in a period is a Group IA atom and the smallest is a noble-gas atom.
- Within each group (vertical column), the atomic radius tends to increase with the period number.

Atomic Radius



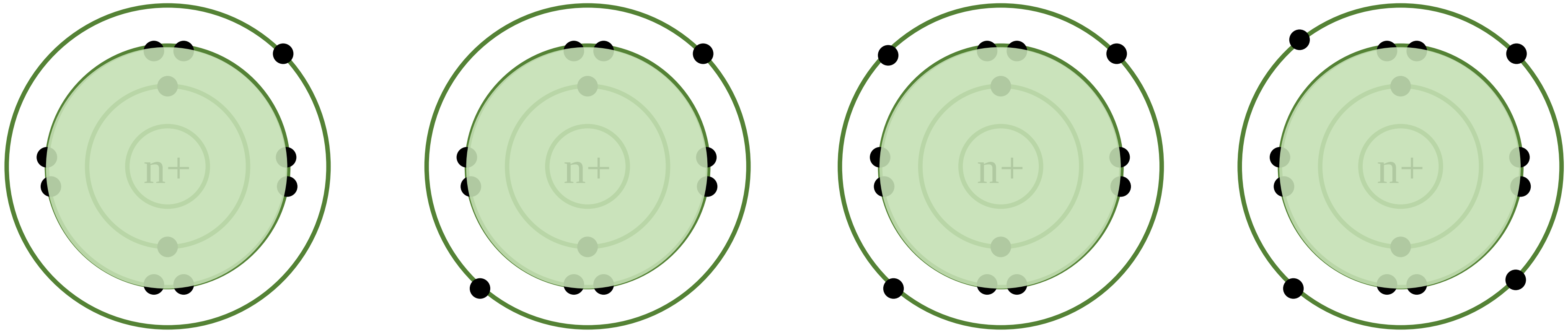
Atomic Radius

There are two factors that primarily determine the size of the outermost orbital

- One factor is the principal quantum number *n* of the orbital; the larger *n* is, the larger the size of the orbital.
- The other factor is the effective nuclear charge acting on an electron in the orbital; increasing the effective nuclear charge reduces the size of the orbital by pulling the electrons inward.

The **effective nuclear charge** is the positive charge that an electron experiences from the nucleus, equal to the nuclear charge but reduced by any shielding or screening from any intervening electron distribution.

Effective nuclear charge= Nuclear charge - Shielding effect



Atomic Radius

Consider a given period of elements. The principal quantum number of the outer orbitals remains constant. However, the effective nuclear charge increases, because the nuclear charge increases and the number of core electrons remains constant. Consequently, the size of the outermost orbital and, therefore, the radius of the atom decrease with increasing Z in any period.

Now consider a given column of elements. The effective nuclear charge remains nearly constant (approximately equal to e times the number of valence electrons), but n gets larger. You observe that the atomic radius increases.

Ionization Energy

The minimum amount of energy required to remove the outermost electron from an atom in the gaseous state is called the **ionization energy**.

$$Al(g) \longrightarrow Al^{+}(g) + e^{-} \qquad I_1 = 580 \text{ kJ/mol}$$

Second ionization energy (I_2): amount of energy required to remove a second electron from the gaseous monopositive cation to form dipositive cation.

$$Al^{+}(g) \longrightarrow Al^{2+}(g) + e^{-} \qquad I_2 = 1815 \text{ kJ/mol}$$

- Ionization energies are usually expressed in electron volts (eV) per atom or in kilojoules per mol (kJ/mol)
- Value of each ionization energy will increase with each removed electron, since the attractive influence of the nucleus increases and will and will require more energy for the removal of an electron from more positive charges.
- Ionization energies measure how tightly electrons are bound to atoms. Low energies indicate ease of removal of electrons and vice versa.

Factors Affecting Ionization Energy

1. Effective nuclear charge
2. Atomic size i.e. atomic radius
3. Principle quantum number
4. Shielding effect
5. Half filled and filled orbitals
6. Nature of orbitals

Effective Nuclear Charge ↑ Ip ↑

Greater the magnitude of effective nuclear charge, higher is the amount of energy needed to remove the outermost shell electron. Thus with the increase of the magnitude of effective nuclear charge, the magnitude of ionization potential also increases. The effective nuclear charge increases from left to right in a period.

Factors Affecting Ionization Energy

Atomic size

Greater is the atomic size of an atom, more far is the outermost shell electron from the nucleus and hence lesser will be the force of attraction exerted by the nucleus on the outermost shell electron. Thus higher the value of atomic radius of an atom, lower will be the ionization energy.

Principal Quantum Number (n)

The greater the value of n for the valence shell electron of an atom, the smaller the distance of this electron from the nucleus and hence lesser will be the force of attraction exerted by the nucleus on it so lesser energy will be required to remove the valence shell electron. Thus with the increase of the principal quantum number of the orbital from which the electron is to be removed, the magnitude of ionization potential increases.

Shielding Affect

The magnitude of shielding effect determines the magnitude of the force of attraction between the nucleus and the valence-shell electron. Greater is the magnitude of shielding effect working on the valence shell electron. The lesser the ionization potential.

Factors Affecting Ionization Energy

Half-filled and filled orbitals

According to Hund's rule, half-filled (ns1, np3, nd5) or completely-filled (ns2. ns6, nd10) orbitals are comparatively more stable and hence more energy is needed to remove an electron from such orbitals. Thus the ionization potential of an atom having half-filled or filled orbitals in its electronic configuration is relatively higher than that expected normally from its position in the periodic table.

Nature of Orbitals

The nature of orbitals of the valence-shell from which the electron is to be removed also influences the magnitude of ionization potential. The relative order of energy of s, p, d and f orbitals of a given nth shell is as:

$$ns < np < nd < nf$$

This order clearly shows that to remove an electron from f-orbital will be easiest while to remove the same from s-orbital will be the most difficult.

Trends in Ionization Energy

- Ionization energy generally increases from left to right in a period because of the increase in effective nuclear charge and decrease in atomic radius.
- Ionization energies generally decrease down a group due to the shielding effect and increase in atomic size.
- Deviation from these trends can usually be occurred due to repulsion between electrons, especially electrons occupying the same orbitals.
  - A IIIA element (ns^2np^1) has smaller ionization energy than the preceding IIA element (ns^2). Apparently, the *np* electron of the IIIA element is more easily removed than one of the *ns* electrons of the preceding IIA element.
  - A VIA element (ns^2np^4) has smaller ionization energy than the preceding VA element (ns^2np^3). As a result of electron repulsion, it is easier to remove an electron from the doubly occupied *np* orbital of the VIA element than from a singly occupied orbital of the preceding VA element.

Electron Affinity

The electron affinity is the energy change for the process of adding an electron to a neutral atom in the gaseous state to form a negative ion.

$$X(g) + e^- \longrightarrow X^- (g)$$

Factors Affecting Electron Affinity

Nuclear charge
More the nuclear charge of the atom more strongly will it attract additional electron. Therefore, electron affinity increases as the nuclear charge increases.

Atomic size
The smaller the size of atom smaller will be the distance between the extra electron and the nucleus. Therefore, electrostatic force of attraction will be more and the electron affinity will be higher.

Electronic Configuration
Atoms having stable electronic configuration (i.e. those having filled or half filled outer orbitals) do not show much tendency to add extra electron, so have either zero or very low electron affinities.

Electron Affinity

- All of the Group IA elements have small negative electron affinities. When you add an electron to a lithium atom, for example, it goes into the 2s orbital to form a moderately stable negative ion, releasing energy.

  Li(1s22s1) + e^- → Li^-(1s22s2); E.A. = -60 kJ/mol

- None of the Group IIA elements form stable negative ions; that is, the electron affinities of these elements are positive. Each of the atoms has a filled ns subshell, so that if you were to add an electron, it would have to go into the next higher energy subshell (np). Instead of releasing energy, the atom would absorb energy.

  For example,

  Be(1s22s2) + e^- → Be^- (1s22s22p1); E.A. > 0

  (For a similar reason, the Group VIIIA elements also do not form stable negative ions.)

- From Group IIIA to Group VIIA, the added electron goes into the np subshell of the valence shell. With the exception of the Group VA elements, the electron affinities tend toward more negative values as you progress to the right through these elements in any period (more energy is released and stabler negative ions are formed). The electron affinity of a Group VA element is generally less negative than the preceding Group IVA element.

Electron Affinity

$$B(1s^2 2s^2 2p^1) + e^- \longrightarrow B^-(1s^2 2s^2 2p^2);\ E.A. = -27 \text{ kJ/mol}$$
$$C(1s^2 2s^2 2p^2) + e^- \longrightarrow C^-(1s^2 2s^2 2p^3);\ E.A. = -122 \text{ kJ/mol}$$
$$N(1s^2 2s^2 2p^3) + e^- \longrightarrow N^-(1s^2 2s^2 2p^4);\ E.A. = 0 \text{ kJ/mol}$$
$$O(1s^2 2s^2 2p^4) + e^- \longrightarrow O^-(1s^2 2s^2 2p^5);\ E.A. = -141 \text{ kJ/mol}$$
$$F(1s^2 2s^2 2p^5) + e^- \longrightarrow F^-(1s^2 2s^2 2p^6);\ E.A. = -328 \text{ kJ/mol}$$